Ananya Nandy

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Education

University of California, Berkeley — *Berkeley, CA*
Ph.D. in Mechanical Engineering — *August 2019 – Exp. 2024*

- Recipient of Departmental First-Year Fellowship

Massachusetts Institute of Technology (MIT) — *Cambridge, MA*
B.S. in Mechanical Engineering (GPA: 4.9/5.0) — *September 2015 – June 2019*

- Thesis: Fabrication Process and Electromechanical Characterization of a Sliding Connector System for Modular Stretchable Electronics
- Tau Beta Pi Engineering Honor Society, Pi Tau Sigma Mechanical Engineering Honor Society

Research Experience

University of California, Berkeley — *Berkeley, CA*
Ph.D. Student — *August 2019 – Present*

- Research assistant in the group of Professor Kosa Goucher-Lambert.
- Investigating functional modeling for human-centered computational approaches to engineering design.

Busch Vacuum Pumps and Systems, R&D Dept. — *Maulburg, Baden-Württemburg, Germany*
Research Intern — *June 2019 – August 2019*

- Designed and executed experiment to characterize acoustics of claw compressor.
- Developed MATLAB tool to determine acoustic impedance.
- Incorporated acoustic impedance to improve performance of silencer acoustic simulation.
- Presented results to company founder, managing partner, and global R&D manager.

MIT Media Lab, Responsive Environments Group (Prof. Joseph Paradiso) — *Cambridge, MA*
Undergraduate Researcher — *September 2018 – May 2019*

- Manufactured novel electro-mechanically robust connector systems to connect modules.
- Conducted electrical and mechanical characterization tests on connectors for stretchable electronic sensor module (part of a modular wearable).
- Wrote senior thesis on results.

MIT Mechanical Engineering Dept., Varanasi Lab (Prof. Kripa Varanasi) — *Cambridge, MA*
Undergraduate Researcher — *February 2018 – May 2018*

- Designed testing devices for microfluidic co-flow in syringes and needles with an application to vaccine delivery.
- 3D printed molds and fabricated the microfluidic devices with PDMS.
- Performed pressure and flow rate experiments with co-flow devices and emulsions.

Mitsubishi Electric R&D Center, Solution Engineering Dept. — *Amagasaki, Hyogo, Japan*
Smart Systems Group Research Intern — *June 2017 – August 2017*

- Applied machine learning methods in Python to analyze energy data for consumer and supplier energy management within smart systems.
- Successfully disaggregated appliance level data from smart meter data.

MIT International Design Center, Urban Risk Lab (Prof. Miho Mazereeuw) *Cambridge, MA*
Undergraduate Researcher *February 2016 – December 2017*

- Increased durability of pedal powered DC generator system components in the PREPHub, a community hub infrastructure designed for post-disaster resilience and daily use.
- Utilized CAD to model units that could be customized for location.
- Built nodes with wireless radio parts and set up Raspberry Pi with CATAN, an emergency mesh communication system developed by MIT Lincoln Lab.

Work Experience

Sistine Solar, Greentown Labs *Somerville, MA*
Mechanical Engineering Intern *June 2018 – August 2018*

- Headed product development process for device to efficiently apply thin SolarSkin films to solar panels.
- Executed design, rapid prototyping, material procurement, manufacturing, and testing for first prototype from concept to completion.
- Worked within a small, fast-paced team and delivered the product with a plan for manufacturing at scale.

Publications and Presentations

Invited Poster

Nandy, A. "Effect of Figure Skating Jump Entry on Jump Quality", MIT Sports Technology Summit. Cambridge, 2018

Selected Projects

Sentry: A Dynamic Speed Bump System for Emergency Response *Cambridge, MA*
Surface Task Force Lead *September 2018 – December 2018*

- Led sub-team of 5 to develop road-surface elements of speed bump.
- Designed, sourced, and manufactured dynamic hinged steel plates and durable rubber surface.
- Performed mechanical load bearing analysis to determine lifting mechanism (supporting car weight).

Motorized Retractable Joystick *Dorchester, MA*
Principles & Practices of Assistive Technology: Mechanical Lead *September 2017 – May 2018*

- Worked in a team to develop an automated retracting joystick for a power wheelchair user in The Boston Home.
- Designed and manufactured the mechanical interfacing for the project.
- Maintained communication with client and organized testing sessions.
- Created detailed and open-source documentation that was featured by the publication website (https://www.instructables.com/id/Developing-a-Motorized-Retractable-Joystick/).

Exploration of Universal Design *Cambridge, MA*
Design for America: Project Team Member *September 2016 – May 2017*

- Interviewed clients and researched assistive technology through universal design.
- Collaborated with a power wheelchair user without fine-finger dexterity to find a solution so he could open packages.
- Designed, manufactured, and delivered device to the user based on human-centered design principles.

Extracurricular Activities and Outreach

Science Policy Group at Berkeley *Present*
- Engaging in science communication through op-eds (currently: plastics and sustainable products).

MIT Maker Workshop Mentor *September 2018 – May 2019*
- Trained students on waterjet and maintained machine.
- Supervised and mentored users at student-run makerspace.

Lab Assistant for Design and Manufacturing II *February 2019 – May 2019*
- Assisted lab section of 10 students with CAD, CAM, and manufacturing for mass-manufactured yo-yo project.

MIT Global Teaching Labs *January 2018*
- Developed hands-on Arduino workshops to introduce engineering to high school students.
- Taught physics and the Arduino workshops to over 200 3rd, 4th, and 5th year students at Leonardo da Vinci Scientific High School in Milan, Italy.

MIT Design for America Project Director *May 2017 – May 2018*
- Led 5 teams through yearlong user-centered design projects with the topics of reuse on-campus, indoor air pollution, storage for the homeless, snow removal in Boston, and women in STEM.
- Developed timelines and design process lessons.
- Organized semester design reviews

Captain of MIT Figure Skating Team *August 2016 – May 2018*
- Coordinated logistics for team of 5 – 10 for intercollegiate competitions.
- Organized team social events and developed team spirit activities.

Vice President of MIT Figure Skating Club *August 2016 – May 2019*
- Assisted with weekly group lessons for >50 members.
- Ran weekly club logistics such as skate rentals as well as helping organize semester-end shows

Peer Mentor at Maseeh Hall *August 2016 – May 2017*
- Advised 20 first-year MIT students in residence hall.
- Implemented events and communicated resources to ensure a smooth first-year transition.
- Mediated conflicts and problems for students.

Organization Committee for MIT Energy Conference *October 2015 – March 2016*
- Contacted and maintained relations with 10 companies for energy showcase.
- Developed and advertised the event with marketing and communications team.

Skills

SolidWorks, Autodesk Fusion 360, MATLAB, Python (Anaconda & Keras), R, CNC/Machining (Mill & Lathe), Waterjet, Laser Cutter, 3D Printing, Electronics/Arduino, Wet Lab Experimentation, Instron Tensile Testing, Microsoft Office

Coursework

User Interface Design, Designing for Emerging Technologies, Introduction to Robotics, Neural Control of Movement, Machine Learning, Mechanics/Materials, Dynamics/Controls, Thermal Fluids Engineering, Measurement/Instrumentation, Design/ Manufacturing, Design for the Developing World, Principles and Practices of Assistive Technology, Electronics for Mechanical Systems, Numerical Computation, Differential Equations, Multivariable Calculus, Physics (Mechanics / Electricity & Magnetism)